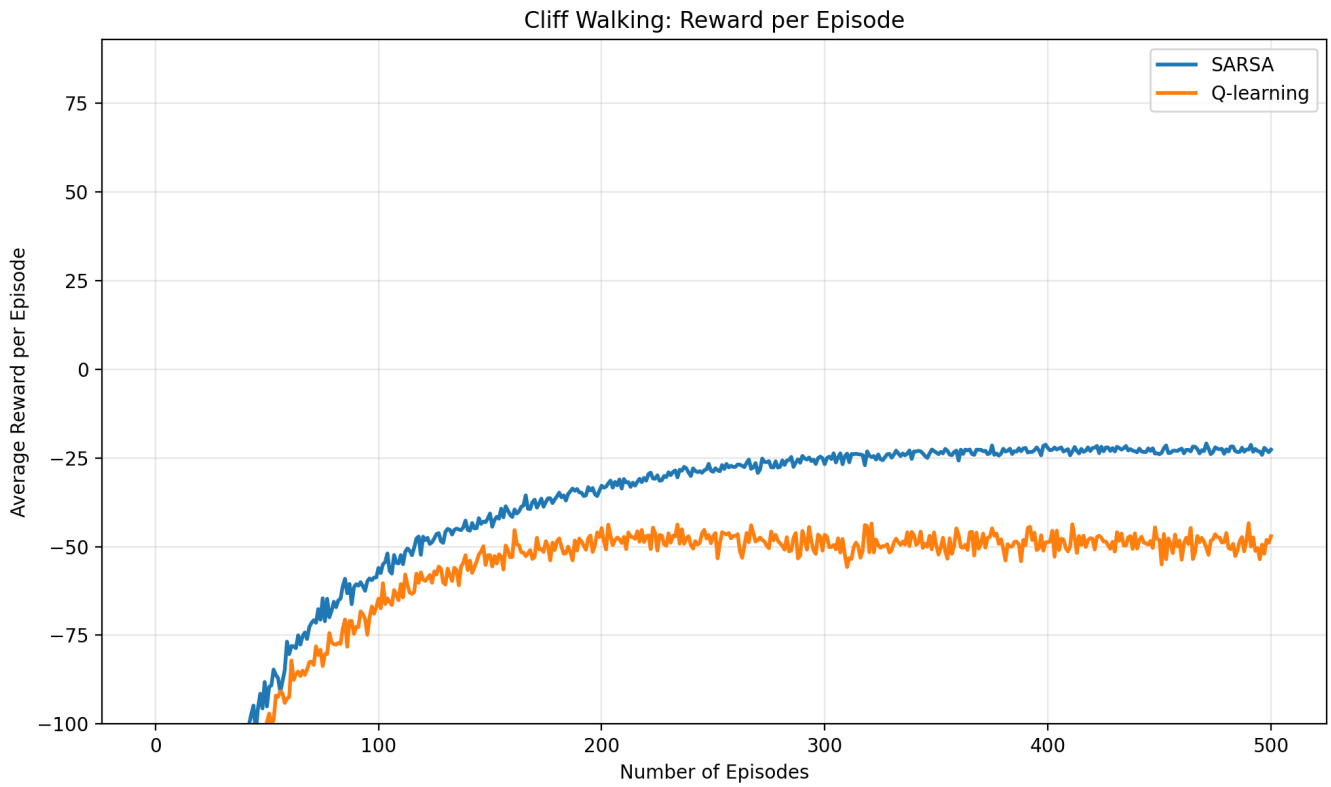
Assignment 4

Q1

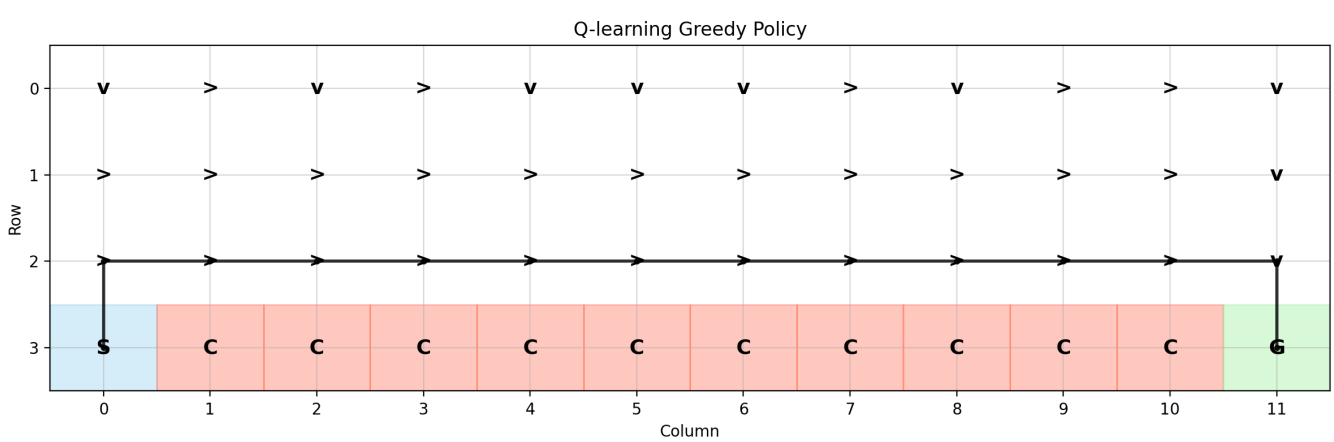
(a)

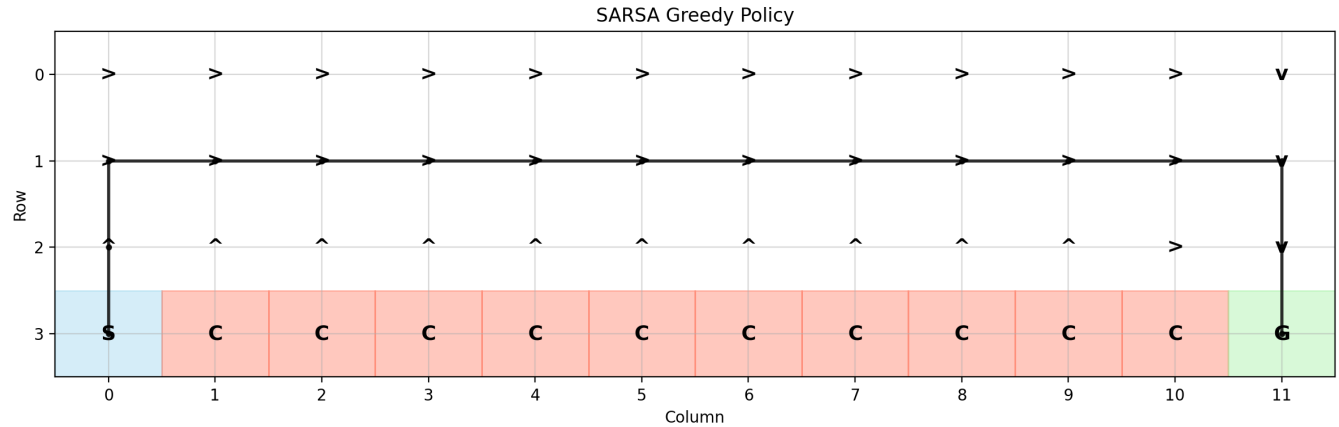
The curve in the textbook is smoother since its most likely average across a lot of runs, the plot below is averaged across 1000 runs



(b)

Q learning plot is the same as in textbook. SARSA is slightly different



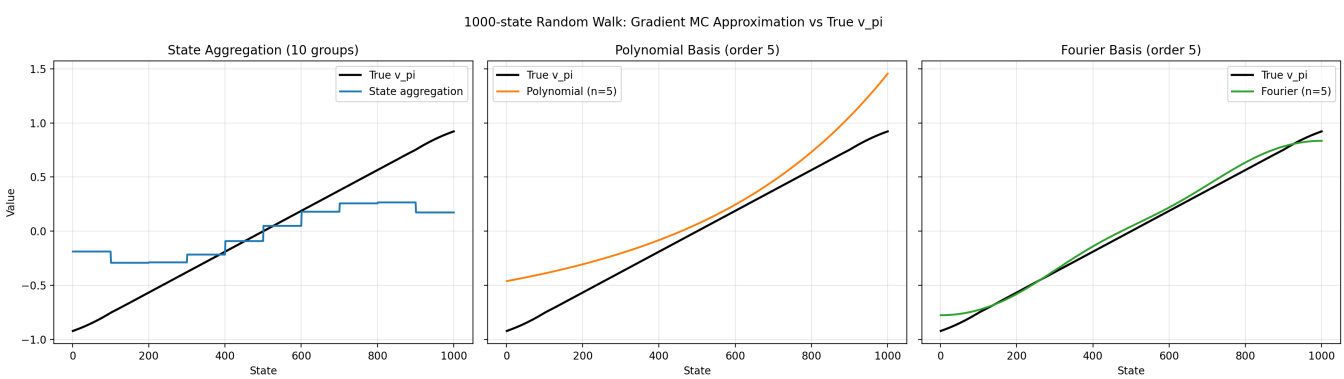


Its possible the epsilon greedy part can give us 2 branches at (row 1, col 0) where it can go up or right, in this case right was chosen and maybe in textbooks case up was chosen. Also here the optimal policy was chosen by taking the average, whereas if take the last iteration alone we could potentially get the textbook's optimal policy

(c)

Based on raw rewards, SARSA gives us better average rewards purely based on the fact it avoids the cliff entirely. But this is only in cliff walk scenario, for other scenarios Q learning being greedy should work better. The best way to experiment would be to see how many times Q learning falls over the cliff compared to SARSA, if that number is small it could be better to use Q learning, and moreover we can try tuning the cliff rewards and see how that affects the rewards graph

Q2



(a)

Fourier with n=5 works best, then polynomial of order 5, then aggregation.
The state aggregation is very simple and stable to do, but as a tradeoff we get a worser fit, as we move on to the polynomial basis, we get a better fit at the tradeoff of runtime and also it becomes sensitive to the data points and parameter tuning. Finally fourier provides the best fit.

(b)

Having a higher order may induce more curvature as is the case here making polynomial not the best option. Finding the optimal order for polynomial is a challenge in itself (hyperparameter tuning), and as order increases it becomes less stable.
I also read online that there is this "Runge's Phenomenon" which basically explains this, when we have a set of equidistant points are trying to fit a higher order polynomial, the curve just gets wobblier and worser as the order increases.